

Customer Churn Analysis for Telecom Industry

Introduction

Customer churn is one of the biggest challenges faced by telecom companies. Losing existing customers affects revenue, increases customer acquisition costs, and reduces long-term business growth. This project aims to analyze customer data, identify the major factors responsible for customer churn, and build a machine learning model that predicts whether a customer is likely to leave the company.

Abstract

This project uses the Telco Customer Churn dataset to analyze customer behavior and predict churn using machine learning techniques. The dataset was cleaned, explored, and preprocessed before training a Decision Tree Classification model. Exploratory Data Analysis (EDA) was performed to identify important trends and relationships between customer attributes and churn. The project also provides business recommendations to improve customer retention.

Objective

- Predict customer churn using machine learning.
 - Identify the key factors influencing customer churn.
 - Analyze customer behavior using data visualization.
 - Provide actionable recommendations to improve customer retention.
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Dataset Information

- **Dataset Name:** Telco Customer Churn Dataset
- **Total Records:** 7,032 (after data cleaning)
- **Total Features:** 21
- **Target Variable:** Churn (Yes/No)

Tools and Technologies Used

- Python
 - Jupyter Notebook
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Scikit-learn
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Methodology

1. Data Collection

The Telco Customer Churn dataset was imported into Jupyter Notebook for analysis.

2. Data Cleaning

- Removed missing values.
- Converted **TotalCharges** into numeric format.
- Removed invalid records.
- Verified data quality.

3. Exploratory Data Analysis (EDA)

Several visualizations were created to understand customer behavior, including:

- Churn Distribution
- Contract Type vs Churn
- Internet Service vs Churn
- Payment Method vs Churn
- Monthly Charges vs Churn
- Tenure vs Churn

4. Data Preprocessing

- Encoded categorical variables using Label Encoding.
- Split the dataset into training and testing sets (80:20).

5. Machine Learning Model

A **Decision Tree Classifier** was used to predict customer churn.

6. Model Evaluation

The model was evaluated using:

- Accuracy Score
 - Confusion Matrix
 - Classification Report
 - Feature Importance
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Key Findings

- Customers with **Month-to-Month contracts** have the highest churn rate.
 - Customers with **low tenure** are more likely to leave the company.
 - **High monthly charges** increase the probability of churn.
 - Customers using **Electronic Check** as the payment method show higher churn.
 - **Contract Type, Tenure, and Monthly Charges** are among the most important factors influencing churn.
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Customer Segmentation

Loyal Customers

- Long tenure
- Stable contract
- Low probability of churn

At-Risk Customers

- Month-to-Month contract
- High monthly charges
- Higher probability of churn

Dormant Customers

- Low tenure

- Less engagement
 - Potential future churn
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Business Recommendations

1. Encourage customers to switch to long-term contracts.
 2. Offer discounts and loyalty rewards to high-risk customers.
 3. Improve customer support and complaint resolution.
 4. Provide personalized retention offers.
 5. Monitor new customers during the first few months of service.
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Conclusion

This project successfully analyzed customer churn using machine learning techniques. The Decision Tree model helped identify important factors responsible for customer churn. The insights generated from this analysis can help telecom companies improve customer retention, reduce churn, and make better business decisions through data-driven strategies.

Future Scope

- Compare multiple machine learning models such as Random Forest and XGBoost.
- Improve prediction accuracy through feature engineering.
- Deploy the model as a web application.
- Create an interactive Power BI dashboard for business users.